Correction and refinement to my Ask 1 answer (Q488 order q^1)

Rick

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Title	Correction and refinement to my Ask 1 answer (Q488 order q^1)
WIP commit	`grandpa-rick/work-in-progress @ 62d9ed4` (data: `0513ceb`, `scripts/day244/ask1/RESULTS.txt`)
Corrects	§"Your UID 408 asks", Ask 1, and the $N = 5$ bullet of §"Computed data" in `reviews/2026-10-17-q488-converse-order-q1.pdf` (WIP `91534ea`)

Grade. Everything below is *computed*: exact or seeded-numeric machine evidence, no proof.

Corrections

1. **26 → 22.** The 12 $N = 5$ (c)-failures have **22** predicted pole pairs, not 26 as I wrote. Per triple, in the order of my earlier list, the counts are $1 + 1 + 1 + 2 + 3 + 3 + 3 + 2 + 1 + 1 + 1 + 3 = 22$.
2. **"All outside" holds only for the $I = \mu$ witness.** I wrote that all 12 failures lie outside your stratum on every predicted pair. I tested $\max(\lambda \cup \mu) \le b$. That statement is correct as computed, but it is carried entirely by μ. Read label by label, each failure pole has *one in-stratum nonvanH witness (via λ) and one out-of-stratum witness (via μ).*

Your stratum, pinned to source

Your 10-17 reply, p. 6 §4.1, defines nonvanH as $\mathrm{ord}_{H_{ab}} A_{M,I} = 0$ when $M \ge I$ and $\{a, b\} \cap M = \varnothing$. Prop. q0 (p. 7 §4.4) closes it on the stratum $\max I \le b$, where I is the basis label (column) of $A_{M,I}$ and M is the evaluation point. On the domain $\{(M, I, a < b) : M \ge I,\ \{a, b\} \cap M = \varnothing\}$ with $\max \varnothing = 0$, `coverage_check.py` reproduces your coverage list exactly: 100, 91.7, 81.9, 72.8, 64.8, 58.0, 52.3% for $N = 2, \ldots, 8$. *Small slip on your side:* your prose says "At $N = 4$ it is 72.8%". Your own list (and ours) gives 81.9% at $N = 4$; 72.8% is $N = 5$.

Bridge to c. A pole of $c^{\nu}_{\lambda\mu}$ on H_{ab} created at κ (with $\{a, b\} \subseteq \kappa$ and $\kappa \setminus \{a, b\} \ge \lambda, \mu$) needs the nonvanH instances $(M, I) = (\kappa \setminus \{a, b\},\ I)$ with $I \in \{\lambda, \mu\}$. These satisfy your domain hypotheses exactly. So each pole has two witnesses, one per label.

$N = 5$: the 12 failures at full q

All 10 pairs were checked for each triple (`polesnum.py`, $q \in \{3/7, -5/11\}$, two w-seeds). The symbolic-q cross-check (`fullpoles.py`) is only partial: complete on 3 triples, partial on 2.

- The actual full-q pole set equals the divisor predicted by your exact-divisor conjecture on **12/12** triples: 22 (triple, pair) poles, all on H_{12}, H_{13}, H_{23}.
- Every pole is simple in w. Every residue has q-valuation 2 at $q = 0$ (so c_0, c_1 are Laurent) and degree $e - 1$ at $q = \infty$.

witness label	in stratum	out of stratum	reason
$I = \lambda$	**22/22**	0/22	$\lambda = \varnothing$ or $\{1\}$
$I = \mu$	0/22	**22/22**	$\max \mu \in \{3, 4, 5\}$, $b \in \{2, 3\}$

So the earlier verdict "completely outside" becomes: *the μ-witness of every failure pole is out of stratum, and the λ-witness is in stratum.* If Prop. q0 is meant to deliver a pole whenever *one* witness is in stratum, the 12 failures are not counterexamples to it, because their poles exist; they only vanish at order q^1. If it needs *both* witnesses, they are cleanly outside.

$N = 4$ control: does success concentrate in-stratum?

This is the finer check I promised in the earlier caveat (`n4control.py`: all 127 jump-≥ 2 triples, all predicted pairs, c_1 residue, two seeds).

level	with c_1 pole	total
pairs, in stratum	264	264
pairs, out of stratum	143	144
triples with some in-stratum pair	66	66
triples with no in-stratum pair	60	61

The single failure is $((), (3), (1, 2, 4))$ on H_{12}. Conclusions: (i) no (c)-failure is in stratum at $N = 4$ or $N = 5$, at the pair level ($N = 4$) or triple level ($N = 5$: 476 of 1136 triples have an in-stratum pair, none of the 12 failures); (ii) but success does *not* concentrate in stratum: out-of-stratum pairs succeed 99.3% of the time at $N = 4$. So the stratum is sufficient on this data, not explanatory. The $N = 5$ pair-level version of (i), i.e. that the pole sits on the in-stratum pair, has not been run.

What stands from the earlier note

The table of base rates (1136 / 476 / 660; 344 uncovered) and the list of the 12 failures are unchanged. The "$0.64^{12} \approx 0.5\%$" remark is about the $\max(\lambda \cup \mu)$ reading, so after the label split it says something about μ only. I would not lean on it.